

The OIA Stack, with *NIST & MITRE* reference mapping

Ten layers, seven cross-layer spans, and how the OIA composes with the frameworks the enterprise already uses — *NIST CSF*, *NIST AI RMF*, *MITRE ATLAS*, *MITRE D3FEND*, *OWASP Top 10 for LLM Applications*, *Confidential Computing Consortium*, and *ISO/IEC 42001*. The OIA is additive to these frameworks, not competitive with them.

L	LAYER · PURPOSE	FRAMEWORK REFERENCES	SPAN CONCENTRATION
9	Human and Browser Interface — <i>surface at which humans and intelligent systems interact</i> Where humans meet AI: prompts, conversational interfaces, browser-as-agent-runtime. Where the named threat surface of mimicry first lands.	NIST CSF · PR / DE MITRE ATLAS MITRE D3FEND	● Security ● Identity ● Auditability
8	Continuity Fabric — <i>architectural centrepiece</i> Cognitive state preserved through change. Recovery, rollback, post-incident integrity for autonomous systems. Witness chains and verifiable continuity.	NIST AI RMF · Manage ISO/IEC 42001	● Security ● Verifiability ● Auditability ● Identity ● Provenance
7	Orchestration and Workflow — <i>agent action, tool invocation, multi-step workflows</i> Where autonomy lives. MCP, agent frameworks, tool registries. Origin or propagation for three of four named threat surfaces (mimicry, agent-in-the-middle, agent spoofing).	MITRE ATLAS · LLM TTPs NIST AI RMF MITRE D3FEND	● Security ● Auditability ● Identity ● Provenance
6	Context and Knowledge — <i>retrieval, grounding, continuous perception</i> Retrieval-augmented generation, agent context windows, prompt injection threat surface. Adversarial pressure on inputs.	OWASP LLM · 01, 03 MITRE ATLAS	● Security ● Auditability ● Provenance
5	Inference and Retrieval — <i>model execution + continuous trust scoring</i> Where models run. Where the named threat surface of model compromise originates. The model is an actor now — weight swap, alignment drift, jailbreak emergence.	OWASP LLM Top 10 NIST AI RMF CCC · Attested Inference MITRE ATLAS	● Security ● Verifiability ● Auditability ● Provenance
4	Model Training and Adaptation — <i>pre-training, fine-tuning, RLHF</i> Where model weights are created and changed. Training data provenance, attested fine-tuning for regulated adaptation.	NIST AI RMF · Govern CCC · Attested training ISO/IEC 42001	● Security ● Verifiability ● Provenance
3	Agent Data Substrate — <i>persistent, auditable, access-controlled data</i> The data environment intelligent systems operate on. AI-Inside problem: embedded foundation models in SaaS create data egress paths traditional DLP wasn't built to detect.	NIST CSF · PR / DE NIST AI RMF ISO/IEC 42001	● Security ● Sovereignty ● Identity ● Auditability ● Provenance
2	Sovereign Infrastructure — <i>compute, storage, networking with sovereignty made explicit</i> Where workloads physically run and under what jurisdictional regime. Three dimensions of sovereignty: jurisdictional, operational, energy.	ISO/IEC 42001 CCC · Attested compute	● Security ● Sovereignty ● Energy ● Verifiability
1	Silicon Abstraction — <i>GPU/TPU/edge runtimes, TEE primitives</i> Hardware diversity behind portable runtimes. Anchor for the Verifiability span — Intel TDX, AMD SEV-SNP, NVIDIA Confidential GPU, ARM CCA.	CCC · TEE primitives	● Security ● Energy ● Verifiability
0 SUBSTRATE	Physical Compute — <i>semiconductors, energy, facilities</i> The physical substrate beneath the architectural stack. Supply chain, grid capacity, geographic siting.	NIST CSF · ID.SC	● Security ● Sovereignty ● Energy

THE SEVEN CROSS-LAYER SPANS

Security Widest scope; threads through every layer.	Sovereignty Jurisdictional · operational · energy. Layer 2 upward.	Auditability The capacity to reconstruct, after the fact.	Identity Three actors: humans, agents, foundation models.	Energy Architectural, not just operational. Layer 0 origin.	Provenance Lineage of information, models, decisions, actions.	Verifiability Cryptographic proof, not assertion. New in v0.5.
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FRAMEWORK REFERENCE KEY

NIST CSF NIST Cybersecurity Framework — functional categories applied across the stack	NIST AI RMF NIST AI Risk Management Framework — Govern · Map · Measure · Manage
MITRE ATLAS Adversarial Threat Landscape for AI Systems — attack TTPs	MITRE D3FEND Defender-side techniques taxonomy
OWASP LLM OWASP Top 10 for LLM Applications — risk enumeration	CCC Confidential Computing Consortium — hardware-attested execution
ISO/IEC 42001 AI management system standard	

The OIA Reference Model is additive to the existing reference architectures and management frameworks the enterprise already uses. Where layers concentrate threat surface (5, 7, 8), framework anchors include MITRE ATLAS and OWASP LLM. Where layers concentrate verifiability (1, 2, 4, 5, 8), the Confidential Computing Consortium provides the foundational primitives. The OIA's contribution is the structural map within which these frameworks compose; without it, the frameworks address overlapping concerns through incompatible vocabularies.